

Anesthesia and Analgesia for Pectus Excavatum Surgery

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KEYWORDS

- Pectus excavatum • Nuss procedure • Thoracic epidural • Haller index
- Chest wall deformities • Multimodal analgesics

KEY POINTS

- Special attention is needed during surgical dissection across the mediastinum due to arrhythmias and potential viscus/vessel perforation.
- Deep extubation is preferable for avoidance of coughing and straining on the endotracheal tube, which can cause subcutaneous emphysema.
- Pain tends to be worse with increased age, because patients with ossified ribcages tolerate the procedure less.
- Multimodal analgesic techniques work best including nonnarcotic analgesics, narcotics, regional anesthesia, muscle relaxants, and anxiolytics.
- Aggressive physical therapy, particularly ambulation, improves dynamic pain control and may decrease chronic postoperative pain.

INTRODUCTION

Pectus deformities, mainly pectus excavatum, are the most common chest wall deformity seen by pediatricians and family practitioners.¹ Originally described by Ravitch in 1949, pectus repair was undertaken in the more severe instances to correct the existing deformity and to prevent its progression. It was believed at the time of surgical correction that younger patients had a higher chance for attaining a normal appearance with subsequent growth of the thoracic cage.²

These deformities are caused by defective growth of the sternum and surrounding costal cartilages. Although criticized by some as a cosmetic repair, other evidence supports surgical repair because of the increased likelihood of progressive cardiopulmonary dysfunction over time.

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After need for surgical repair has been established, the technique of choice for the last couple of decades has been the Nuss procedure.³ This is a minimally invasive technique in which rigid metal bars are placed transthoracically beneath the sternum and costal cartilages for a period of time until permanent remodeling of the chest wall has occurred. That length of time tends to be 2 years before bars are removed.

During anesthesia, the three areas of focus include (1) potential for considerable blood loss, (2) potential for lethal arrhythmias, and (3) potential consequences of pneumothoraces. Postoperative pain is considerable, making multimodal analgesia the hallmark of postoperative pain control. Controversy exists regarding the best method of controlling postoperative pain, whether the emphasis is placed on patient-controlled analgesia (PCA) or regional anesthesia. Part of the answer may be found by examining patient outcomes. However, measuring patient outcomes depends on the identity of the enquirer. Surgeons, anesthesiologists, pain services, and hospital administration may all have different end points. Surgeons may look at optimal correction of the deformity followed by a short length of stay. The anesthesia team may be invested in a safe passage through anesthesia without complications. The pain service may focus on optimal postoperative pain control, and the hospital administration may be most concerned with cost and patient satisfaction scores.

EPIDEMIOLOGY

Pectus excavatum occurred in approximately 0.12% of patients in one autopsy series.⁴ Researchers investigated the prevalence of pectus excavatum in teenage public school students in Brazil, and found it to be 1.275%.⁵ Of the congenital chest wall deformities, between 3% and 4% fall under the umbrella of rare deformities (eg, pentalogy of Cantrell, Jeune syndrome, and Poland syndrome).⁶ Pectus excavatum and pectus carinatum represent the remainder, with 90% of those being pectus excavatum.^{1,6}

Gender distribution favors males, noted in different studies from a 2:1 ratio up to 9:1.^{5,6} Although no genetic defect has specifically been discovered, positive family histories, up to 65.5% in one study, make this predisposition likely.⁵

Depending on the surgeon, the type of operation, and the presence of connective tissue disorders, recurrence occurs with unsatisfactory results thus requiring reoperation in 2% to 37%,⁷ although most series claim recurrence rates of 2% to 10%.⁸

PATHOPHYSIOLOGY

The cause of pectus deformity is largely unknown. There are multiple hypotheses regarding the origin. The current dominant theory to explain the development of these two most common chest wall deformities is overgrowth of the costal cartilages.⁹ Collagen type II is one of the major structural components of rib cartilage, but results of histologic and biochemical studies of costal cartilages in patients with pectus excavatum remain inconclusive.⁹

CLINICAL PICTURE

Patients with pectus excavatum, or “funnel chest” as it is known, typically present for repair during adolescence. There has been some discussion in the literature as to the best timing for surgical repair. In the past, these defects were repaired at an earlier age (4–6 years of age) using a Ravitch procedure,² which removes three to four overgrown cartilages and repositions the sternum.¹⁰ Even Nuss tended to repair these defects prepubertally. His patients had an age range of 1 to 15 years old, with most in the

3- to 5-year range.³ This contrasts with more recent studies involving the Nuss procedure, which have a mean age of between 13 and 23 years of age.^{8,11–13}

Pectus repair can be problematic in very young children. Patients may be more active following discharge and have a greater rate of sustaining unexpected trauma.³ Rib growth after early pectus correction can also be impaired in younger patients, sometimes to the point of minimizing chest wall movement, also known as asphyxiating thoracic chondrodystrophy.³

Repairs in older patients can be more difficult, with prolonged operating time and more blood loss.³ The trend in current practice is to wait for patients to reach their middle teenage years to perform a pectus excavatum repair. This allows for the patients to complete their teenage growth spurt and have a less likely chance of recurrence. Nonetheless, older school-age children and younger teenagers with significant cardiorespiratory compromise affecting their ability to participate in aerobic sports could also be candidates for repair.

Patients with connective tissue disorders, such as Marfan syndrome and Ehlers-Danlos syndrome, are noted to have a higher incidence of recurrence.⁷ Therefore, one might consider waiting to correct these deformities. However, this subset of patients is at risk for aortic dilation and aortic valve insufficiency. Patients with normal cardiac anatomy may have repair of their pectus excavatum on an elective basis, but those in need of aortic arch repair may need to have their pectus repaired at the time of open heart surgery, or deferred to a later date after cardiac repair.¹⁴ Thus, the timing of surgery for patients with these connective tissues disorders must also be carefully assessed.

Reasons for postpubertal repair include body image issues that become manifest during puberty, recurrence of the defect, and progressive cardiopulmonary dysfunction. Patients may have dyspnea on exertion, shortness of breath, breathlessness, a feeling of heaviness in the chest, air hunger, and exercise intolerance.¹⁴ These symptoms can be progressive, and may not be eminent until the patient or family notices difficulties in “keeping up with their peers.”

Indications for surgery to correct a pectus excavatum include symptomatic patients, which includes exercise intolerance, decreased endurance, and pain; body image issues; Haller index greater than 3; caliper measurement depth greater than 2.5 cm; exercise-induced asthma; abnormal pulmonary function tests (PFTs); and compression of the right atrium or right ventricle on echocardiogram.¹⁴

Currently, the work-up for pectus excavatum repair includes a history and physical examination and frequently includes a computed tomography (CT) scan of the chest. Since the early 1990s, CT scans have been used to evaluate patients with pectus deformities for surgery. CT scans more clearly document the severity of the deformity, along with the degree of cardiac compression and displacement, the degree of lung compression, and other unexpected problems.³ In patients with known cardiac issues, a cardiac magnetic resonance imaging has been used. Additional work-up may include echocardiography, caliper measurement of the depth of pectus deformity (>2.5 cm),¹⁴ and PFTs. In general, most patients with significant pectus excavatum have low normal or below normal PFTs. Both static and exercise PFTs are useful in eliciting a physiologic effect of the pectus excavatum defect on the patient.

The Haller index is a measure of pectus excavatum severity using a CT scan measurement. The CT is used to demonstrate the abnormal ratio between the patient's transverse diameter and their anterior-posterior diameter.¹⁰ An index of more than three is considered a moderately severe or greater defect.^{10,14} The width of the chest between the ribs at the lowest level of the pectus defect is measured in centimeters and divided by the height of the chest from the anterior aspect of the spine to the

back of the lowest part of the sternal defect. The quotient from this division is the Haller index (Figs. 1 and 2).

An echocardiogram may be useful to perform because it may show the depressed sternum compressing the right atrium and right ventricle, which may interfere with diastolic filling of these structures. Significant compression of the right heart chambers by the depressed sternum is an indication for surgical correction.¹⁴ Mitral valve prolapse is also commonly seen in patients with pectus excavatum.

Patients with pectus excavatum should also be assessed for connective tissue disorders. Less than 1% of patients with pectus excavatum may have an underlying connective tissue disorder, such as Marfan syndrome, Ehlers-Danlos syndrome, or Loeys-Dietz syndrome. If this class of disorders is suspected then further genetic, cardiac, and ophthalmologic evaluations should be performed.¹⁴

Most pectus excavatum defects involve the lower end of the sternum with the turning down of the sternum and inward turning of costal cartilages four through seven. In regards to the Nuss procedure and surgical technique, two bars are more effective than a single bar but may cause overcorrection in some patients. Those with Marfan syndrome and other connective tissue diseases have soft bones and therefore require two bars to distribute the pressures over a wider area.³

The advantages of the minimally invasive technique include no anterior chest wall incision, no need to raise pectoralis muscle flaps, and no need to resect rib cartilages or perform sternal osteotomy. Other advantages include short operating time, minimal blood loss, early return to full activity, and excellent long-term cosmetic results.³ Disadvantages of the Nuss minimally invasive technique include greater postoperative pain; longer inpatient length of stay; higher cost compared with other techniques; and more involved methods of pain control, including regional anesthesia.¹³

ANESTHESIA MANAGEMENT

The anesthesia management of pectus excavatum repairs can be divided into three main categories: (1) preoperative evaluation and preparation, (2) intraoperative management, and (3) postoperative analgesia.

Preoperatively, the degree of cardiac and pulmonary dysfunction is important to assess. This information can be obtained from a thorough history and physical examination focusing on exercise tolerance and symptomatology. Further knowledge of the cardiopulmonary insult can be obtained from the CT scan and Haller index,



Fig. 1. Anterior photo of preoperative pectus patient. Note the incisional scar from original pectus repair at earlier age.



Fig. 2. Photo from more lateral aspect. Pectus deformity more noticeable from this aspect. Note the “rolling” forward of the shoulders and kyphosis, which are not unusual for these patients.

electrocardiogram, echocardiogram, and PFTs. Knowledge about cardiac compromise, arrhythmias, and pectus recurrence can only help to increase situational awareness.

Preparation before the case is essential, and the need to be prepared for potentially large blood loss is foremost. Type and screening of the patient’s blood is necessary to receive emergency blood products in case of catastrophe. It is mandatory to receive confirmation of potential blood products in the operating room before proceeding with surgery.

Decisions concerning methods of postoperative pain control need to be made and addressed preoperatively. Typically, a postoperative pain service manages the patient’s pain after the procedure. The pain service confers with surgery and anesthesia about the type of regional anesthesia and type of narcotic. Regional anesthesia includes anesthesia-driven blocks, such as thoracic epidural and paravertebral blocks; and surgeon-driven blocks, such as intercostal nerve blocks, wound catheter placement, and local infiltration. Anesthesia-driven blocks need to be placed preoperatively and many choose to do awake thoracic blocks when possible. The practitioner cannot use the lateral decubitus position after surgery for fear of dislodging the newly placed pectus bar.

Because this surgery is extremely painful, another important preoperative consideration, preferably days to weeks preoperatively, is to alleviate anxiety among the patient and family, and manage expectations about the surgery. A chief method of diminishing anxiety and creating reasonable expectations is through education. Parents and patients, if possible, are expected to attend a mandatory educational class about the surgery and hospital course, or view a simulcast of the class. This helps to create more appropriate expectations and assuage anxiety. Once the preoperative evaluation and preparations are complete, the patient is ready for the operating room.

Intraoperative management is often as follows. The patient is brought into the operating room, and after placement of monitors, a peripheral intravenous (IV) line is placed with the aid of nitrous sedation. After securing IV access, the patient is lightly sedated and assisted to the sitting position in preparation for placement of regional anesthesia, which frequently consists of a thoracic epidural.

Compliance with inserting the epidural catheter awake may be difficult in more immature teenagers and young children. Typically, this is attempted in adults and teenagers, or uncommonly in less anxious children. After local anesthetic is infiltrated subcutaneously, an 18-G Tuohy needle is inserted between the spinous processes of

T5-6 or T6-7, and the epidural space noted by loss of resistance to air or preservative-free saline. Placement of the epidural at higher intervertebral spaces has been noted postoperatively to have a higher rate of failed block; higher rate of Horner syndrome; and higher rate of upper extremity issues, such as weak grip and numbness. Weak grip and numbness, although confirmatory for success of block, may cause distress for the patient and necessitate changes in rate and concentration of epidural solution. The catheter is advanced 3 to 4 cm into the epidural space and secured in place. Further advancement has been noted to result in one-sided and failed blocks. A test dose is administered, and once confirmed negative a sterile dressing is placed around the catheter and the patient is returned to a supine position. If there are problems placing the regional anesthetic in the seated patient because of compliance or positioning issues, it may be more advantageous to induce anesthesia, secure the airway, and then place the epidural in the lateral decubitus position.

In regards to further access and monitoring, a second, preferably large-bore peripheral intravenous line (PIV) is placed in preparation for possible rapid fluid or blood administration. An arterial line is not required unless the patient has significant cardiac or pulmonary derangements requiring continuous blood pressure monitoring or frequent blood gas measurements.

The patient is initially placed in a supine position; however, because the surgeon needs to enter and exit through the midthorax by way of the anterior to midaxillary line, the patient's arms must be placed in anatomic position, palm out, at right angles above their head. To minimize traction injuries of the upper extremity, padding is important, as is keeping the patient's joints at right angles.

There are some major anesthetic concerns intraoperatively, particularly injury to the heart and other mediastinal structures, that have been described with this procedure. This complication can happen at the time of bar placement or later, when the bar is removed. With placement, use of bilateral thoracoscopy aids in the careful dissection around the lungs, mediastinum, and pericardium to pass to the contralateral pleural space.¹⁵ Mediastinal dissection is typically the time when arrhythmias can be an issue. Usually, the volume of the pulse oximetry monitor is increased to help the surgeons recognize moments that the dissection may be imperiling the patient. Auditory cues and heightened focus of the electrocardiogram at this time are expected.

After the bar (or bars) is in place and secured, management of the iatrogenic bilateral pneumothoraces is important. Typically, temporary suctioning of the pleural spaces along with a Valsalva maneuver is used to alleviate the pneumothorax on either side. An intraoperative chest radiograph is used to confirm this.

Avoidance of coughing or straining is preferable to circumvent the development of subcutaneous emphysema, created by forcible expulsion of residual pneumothorax. Obviating this can be achieved by deep extubation or significant narcotics to avoid straining on the endotracheal tube.

Intraoperative mortality from exsanguination caused by cardiac or vascular injury has been described, and although rare, heightened vigilance, adequate IV access, and availability of packed red blood cells are important. When these patients return for pectus bar removal after several years, occasionally the bar has become adherent to the lung or pericardium resulting in a severe, sudden, and catastrophic rupture of a major vessel or heart chamber on removal.¹⁶ It is important to be prepared and establish ample IV access to massively resuscitate and transfuse fluids to these patients should a catastrophe occur.

Other common risks postoperatively include bar displacement, residual pneumothorax, and infection; however, cardiac injury, sternal erosion, arterial pseudoaneurysm, and persistent cardiac arrhythmias are also possible.

Patients undergoing thoracic surgery experience a great amount of postoperative pain, particularly in the older subset because the ribcage has become more rigid and ossified. This may be caused by surgical retraction and dissection of intercostal muscles, and conformational change of an ossified ribcage. These patients have been shown to have better pain control and improved postsurgical ventilation and pulmonary toilet with epidural analgesia.^{17–20} However, there are studies that show similar pain control between thoracic epidural and PCA.^{21–24} Whichever approach to postoperative pain is chosen, the regimen for pectus excavatum repairs requires a multimodal approach.

Patients and families should be educated preoperatively in regards to the high degree of postoperative pain expected despite being a minimally invasive approach. Classic studies have demonstrated that the knowledgeable patient requires less analgesia in the postoperative period, and at the same time experiences less pain than the less informed patient.²⁵ The transition from an epidural to oral pain medication can also be challenging and often requires dosing or medication adjustments to establish an optimal regimen. Aggressive physical therapy, particularly ambulation, improves dynamic pain control and may decrease chronic postoperative pain.²⁵ It is vital for the patient's care team to encourage movement up to the chair and ambulation even before the epidural has been removed.

Patients who have had previous repair of their pectus excavatum with either a modified Ravitch procedure or a Nuss procedure may present with recurrent disease. For patients with recurrent pectus excavatum it is important to know if the patient is having recurrent symptoms. Patients often may present with pain associated with their recurrence, and a return to their original pulmonary symptomatology. Patients with chronic pain as the only feature of their recurrence may continue to have pain after repair.¹⁴

There are numerous protocols established to manage postoperative pain control after pectus excavatum repairs. Institutions use either a PCA or regional anesthesia, either a thoracic epidural or paravertebral blocks as the main method of controlling immediate postoperative pain.

One such protocol that has been used effectively begins with the placement of a thoracic epidural in the T5-6 or T6-7 interspace followed by a loading dose of local anesthetic (0.2% ropivacaine) up to 10 mL based on weight, given in an incremental fashion. When the loading dose is complete, an epidural infusion is started consisting of local anesthetic plus clonidine. For the local anesthetic in this mixture a higher concentration is often used (eg, 0.15% ropivacaine). The epidural is run until postoperative Day 3 for all pectus repairs.

The multimodal approach to pain management is used and has been shown to be most effective. Intermittent IV narcotics are available to the patient as needed; however, some patients may need a PCA to supplement the epidural analgesia. Other supplemental medications include methocarbamol (Robaxin), 15 mg/kg IV, with a maximum dose of 1000 mg every 8 hours for 3 days followed by transition to oral Robaxin. Diazepam (Valium), 0.05 mg/kg IV every 4 hours as needed, is very effective in the treatment of muscle spasms and postoperative anxiety, and ketorolac (Toradol), 0.5 mg/kg IV with a maximum of 30 mg every 6 hours for 2 days started on postoperative Day 1, is a useful anti-inflammatory agent. IV acetaminophen every 6 hours is started intraoperatively and continued for up to 3 days postoperatively, then transitioned to the oral form as needed.

As the patient begins oral intake, they also begin to be transitioned to oral pain medications. Oxycodone (Oxycontin), 10 mg twice daily, is started on the first postoperative day. This can be titrated upward if needed. Oxycodone, 0.1 to 0.2 mg/kg every 4 hours, is started on the second postoperative day in the morning. The urinary catheter is also removed approximately 4 hours after epidural removal.

All medications are switched to the oral route, and doses and frequency are adjusted as needed to provide optimal patient comfort. The intermittent IV narcotic is kept available for breakthrough pain after epidural removal, because this is a transition when pain scores often increase by one to two points until an optimal oral pain medication regimen has been achieved. Pain medications are continued for 2 weeks postoperatively; older patients may need pain medication coverage for longer. Patients older than the age of 17 years may require narcotic pain medications longer than 2 weeks postoperatively, and patients with excessive pain issues during

Box 1

Cincinnati Children's Hospital pectus excavatum postoperative pain protocol

Preoperative preparation

- Manage expectations: despite “minimally invasive” still quite painful
- Pain plan: agree and consent for regional anesthesia if agreed on

Day of surgery (POD#0)

- Epidural placement (T5-7) by anesthesia/pain team
- Epidural solution: ropivacaine, 0.15%–0.2% with clonidine, 1 µg/mL
- IV acetaminophen (Ofirmev), 15 mg/kg IV (Max 1000 mg) every 6 hours ATC × 3 days
- IV methocarbamol (Robaxin), 15 mg/kg IV (Max 1000 mg) every 8 hours ATC × 3 days
- IV diazepam (Valium), 0.05 mg/kg IV every 4–6 hours prn for spasm
- IV opioids, if needed, prn or PCA

Postoperative Day 1 (POD#1)

- Continue epidural analgesia, IV acetaminophen, IV methocarbamol, IV diazepam prn
- Start IV ketorolac (Toradol), 0.5 mg/kg up to 15 mg IV every 6 hours ATC
- Start oxycontin, 10 or 20 mg PO every 12 hours; if needed, make every 8 hours

Postoperative Day 2 (POD#2)

- Start oxycodone, 0.1–0.2 mg/kg PO every 4 hours; schedule as ATC, but may be held for parent-patient refusal, respiratory depression, oversedation (Ramsey 4 to 6)
- Continue epidural analgesia, IV acetaminophen, IV/PO methocarbamol, IV/PO diazepam as needed
- Place order in electronic records to stop epidural at 6 AM on POD#3

Postoperative Day 3 (POD#3)

- Stop epidural at 6 AM and remove on morning pain rounds
- Discontinue urinary catheter 4 hours after stopping epidural (~10 AM)
- Once epidural is out: continue oxycontin/oxycodone, PO acetaminophen every 6 hours (maximum, 75 mg/kg/d), PO methocarbamol ATC and/or PO diazepam as needed
- Start PO ibuprofen (10 mg/kg/dose, up to 3 times per day)

Postoperative Day 4+ (POD#4+)

- Keep on service and help manage postepidural pain control
- If 17 years+, history of chronic pain, or prolonged need for oral narcotics >2 weeks, have patient follow up in chronic pain clinic to assist in titration off of narcotics

Abbreviations: ATC, around the clock; IV, intravenous; PO, per os (oral); POD, postoperative day.

hospitalization are followed by the chronic pain service until successfully transitioned off pain medications (**Box 1**).

Problems that can be encountered with thoracic epidural analgesia include hypotension, failed epidural, one-sided epidural, upper extremity weakness or tingling, Horner syndrome, inadequate dermatomal coverage requiring a higher epidural rate, and in rare but serious cases an epidural hematoma or an epidural abscess. The epidural site should be checked multiple times daily by the pain management team and nursing staff to assess for signs of infection at the site or dislodgement of the epidural catheter. Fever is also important to follow up, because this can be a sign of epidural infection. Typically, sustained high fever is an indication for epidural removal.

Other forms of regional anesthesia have been described in lieu of placing a thoracic epidural and have been successful in treating postoperative pain control. These include bilateral intercostal nerve blocks, bilateral paravertebral catheters, and wound catheter infusions.

In conclusion, pectus excavatum repairs are a common type of congenital chest wall deformity, and although minimally invasive are categorized as extremely painful procedures. Preoperative evaluation should focus on patient symptoms, cardiopulmonary examination, and radiologic evidence of the magnitude of the defect. Preparation should include patient-parent education, readiness of donor blood products, and establishment of a postoperative pain management plan. Intraoperative vigilance is extremely important during bar placement and dissection because this could be a time of catastrophic events. Postoperatively, thoracic epidurals are the superior form of postoperative pain control, in addition to use of supplemental medications as part of a multimodal analgesic technique.

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